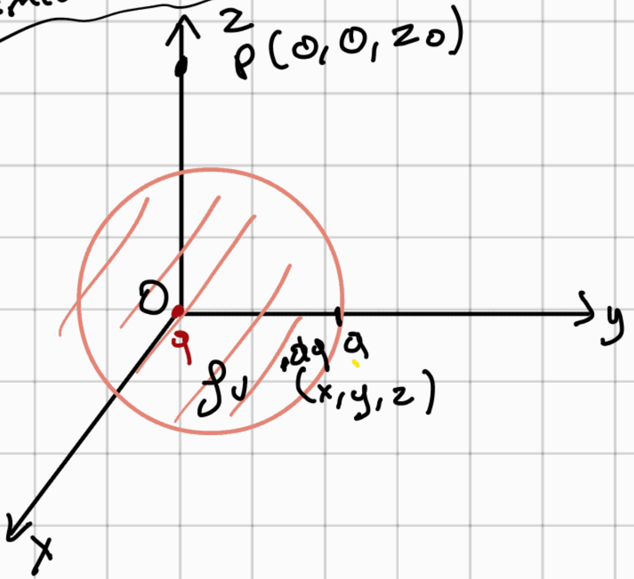
S 17 pr 1

çözümlü sayfa 7



Toplam yük $= q = \int_V \rho_v \, dV = \int_{r=0}^{a}\int_{\phi}^{2\pi}\int_{\theta=0}^{\pi} \rho_v r^2 \sin\theta \, dr \, d\phi \, d\theta$

$dV = r^2 \sin\theta \, dr \, d\theta \, d\phi$

Tek nokta olsaydı

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{qQ}{R^2} \vec{E}_R$$

$x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

İspatlayalım

$$dq = \rho_v \, dV = \rho_v r^2 \sin\theta \, dr \, d\phi \, d\theta$$

$$d\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{dq\,Q}{R^2} \vec{e}_R = \frac{1}{4\pi\varepsilon_0} Q \rho_v r^2 \sin\theta \, dr \, d\theta \, d\phi \, \frac{\vec{R}}{R^3}$$

$$\vec{R} = -x\vec{e}_x - y\vec{e}_y + (z_0 - z)\vec{e}_z$$
$$= -r\sin\theta\cos\phi\,\vec{e}_x - r\sin\theta\sin\phi\,\vec{e}_y + (z_0 - r\cos\theta)\vec{e}_z$$

$$R = \sqrt{r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi + z_0^2 - 2z_0 r\cos\theta + r^2\cos^2\theta}$$
$$= \sqrt{r^2\sin^2\theta + z_0^2 - 2z_0 r\cos\theta + r^2\cos^2\theta} = \sqrt{r^2 + z_0^2 - 2z_0\cos\theta}$$
$$= R$$

$$\vec{F} = \iiint_V d\vec{F} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^{a} \frac{1}{4\pi\varepsilon_0} Q \rho_v r^2 \sin\theta \, dr \, d\theta \, d\phi$$

0 ϕ integrali 0'dan 2π'ye

$$\frac{\vec{R}}{R^3} = \left[\frac{-r\sin\theta\cancelto{0}{\cos\phi}\,\vec{e}_x - r\sin\cancelto{0}{\theta}\sin\phi\,\vec{e}_y + (z_0 - r\cos\theta)\vec{e}_z}{(r^2 + z_0^2 - 2z_0 r\cos\theta)^{3/2}} \right]$$

$$= \int_{\theta=0}^{\pi} \int_{r=0}^{a} \frac{1}{4\pi\varepsilon_0} Q \rho_v r^2 \sin\theta \, 2\pi \frac{(z_0 - r\cos\theta)\vec{e}_z}{(r^2 + z_0^2 - 2z_0 r\cos\theta)^{3/2}} \, dr\, d\theta$$

$$(z_0 - r\cos\theta) = \frac{2z_0^2 - 2z_0 r\cos\theta}{2z_0}$$

$$= \frac{z_0^2 + R^2 - r^2}{2z_0} \rightarrow$$ Allah bilir nasıl oldu

$$= \frac{1}{2z_0}\left[R^2 + (z_0^2 - r^2)\right]$$

$$R^2 = r^2 + z_0^2 - 2 r z_0 \cos\theta$$

$$2R\,dR = 2r z_0 \sin\theta\, d\theta$$

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \frac{2\pi Q \vec{e}_z}{2z_0^2} \int_{r=0}^{a} \rho_v r\,dr \int_{R=z_0-r}^{z_0+r} \left[\frac{z_0^2 + R^2 - r^2}{R^3}\right] R\,dR$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{4\pi Q \vec{e}_z}{z_0^2} \int_{r=0}^{a} \rho_v r^2 dr$$

Neyse yeni soruya geçelim

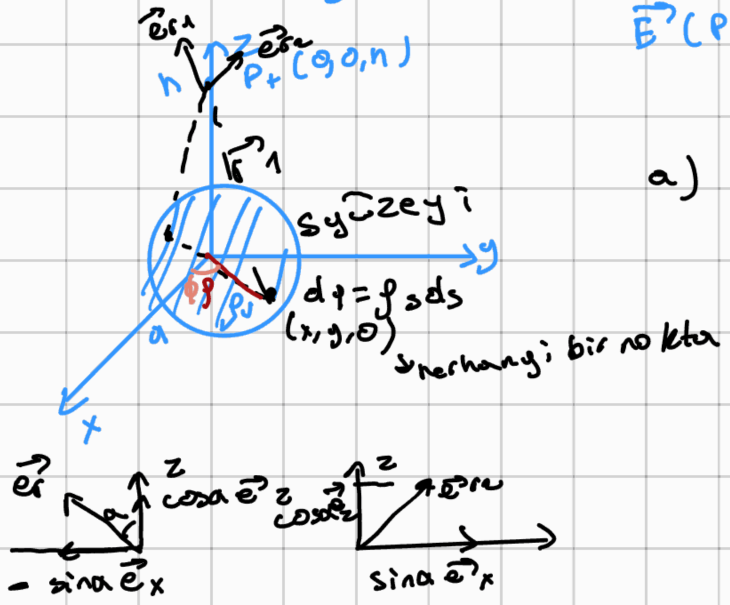
s.19.pr4 mithat idemin

s.22 4.soru elimizdeki baskıda

$x^2 + y^2 \leq a^2$, $z = 0$ ρ_s yük yoğunluğu

$\vec{E}(P_+) = e(h,a)\vec{e}_z$

a)


$\vec{e}_{r1}$
z
$\vec{e}_{r2}$
h
$P_+(0,0,h)$
$\vec{r}_1$
s yüzeyi
y
φ
ρ_s
$dq = \rho_s ds$
$(x,y,0)$
herhangi bir nokta
a
x
$\vec{e}_r$
z
α
$\cos\alpha \vec{e}_z$
$-\sin\alpha \vec{e}_x$
z
$\vec{e}_{r2}$
$\cos\alpha \vec{e}_z$
$\sin\alpha \vec{e}_x$

a) $\vec{e}_{r_1} + \vec{e}_{r_2} = 2\cos\alpha\,\vec{e}_z$

Herhangi binokta için geçerlidir

$dq = \rho_s\, ds = \rho_s\, \rho\, d\rho\, d\phi$

$\vec{r}_1 = -x\,\vec{e}_x - y\,\vec{e}_y + h\,\vec{e}_z$

$x = \rho\cos\phi$

$y = \rho\sin\phi$

$ds = \rho\, d\rho\, d\phi$

$\vec{r}_1 = -\rho\cos\phi\,\vec{e}_x - \rho\sin\phi\,\vec{e}_y + h\,\vec{e}_z$

$r = \sqrt{\rho^2 + h^2}$ //

$$d\vec{E}(P_+) = \frac{dq}{4\pi\varepsilon_0}\frac{\vec{r}_1}{r_1^3} = \frac{\rho_s}{4\pi\varepsilon_0}\frac{\rho\, d\rho\, d\phi}{(\rho^2+h^2)^{3/2}}\left[h\vec{e}_z - \rho\cos\phi\,\vec{e}_x - \rho\sin\phi\,\vec{e}_y\right]$$

$$\vec{E}(P_+) = \int_S dE(P_+) = \frac{\rho_s}{4\pi\varepsilon_0}\int_{\rho=0}^{a}\int_{\phi=0}^{2\pi}\frac{\rho\, d\rho\, d\phi}{(\rho^2+h^2)^{3/2}}\left[h\vec{e}_z - \overset{0}{\cancel{\rho\cos\phi\,\vec{e}_x}} - \overset{0}{\cancel{\rho\sin\phi\,\vec{e}_y}}\right]$$

$$= \frac{\rho_s}{4\pi\varepsilon_0}\left[h\vec{e}_z\int_{\rho=0}^{a}\frac{\rho\, d\rho}{(\rho^2+h^2)^{3/2}}\underbrace{\int_{\phi=0}^{2\pi} d\phi}_{2\pi}\right]$$

$$= \frac{\rho_s h\vec{e}_z}{2\varepsilon_0}\int_{\rho=0}^{a}\frac{\rho\, d\rho}{(\rho^2+h^2)^{3/2}}$$

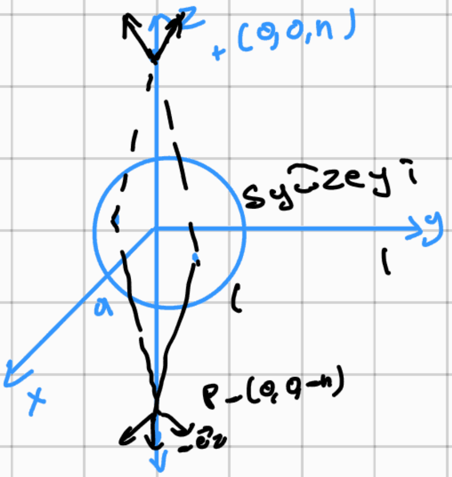
$$\boxed{\int_x \frac{x}{(x^2+a^2)^{3/2}} = \frac{-1}{\sqrt{x^2+a^2}} + C}$$

$$= \frac{\rho_s h\vec{e}_z}{2\varepsilon_0}\left(-\frac{1}{\sqrt{\rho^2+h^2}}\right)_{\rho=0}^{a} =$$

$$= \frac{\rho_s h\vec{e}_z}{2\varepsilon_0}\left(\frac{1}{h} - \frac{1}{\sqrt{a^2+h^2}}\right) = \frac{\rho_s\,\vec{e}_z}{2\varepsilon_0}\left(1 - \frac{h}{\sqrt{a^2+h^2}}\right)$$

$$\Rightarrow \vec{e}(h,a) = \frac{\rho_s\,\vec{e}_z}{2\varepsilon_0}\left(1 - \frac{h}{\sqrt{a^2+h^2}}\right)$$

b)



$\vec{E}(P_-) = -\vec{E}(P_+) = -e(h,a)[-\vec{e}_z]$

$= -e(h,a)\vec{e}_z$

c) $$\lim_{h\to 0} \vec{E}(P_+) = \frac{\rho_s \vec{e}_z}{2\varepsilon_0} \underbrace{\lim_{h\to 0}\left(1-\frac{h}{\sqrt{a^2+h^2}}\right)}_{1}$$

$$= \frac{\rho_s \vec{e}_z}{2\varepsilon_0}$$

daireye yaklaştıkça a ve h bağlantısı azalır 0 olunca yok olur sonsuz düzlem gibi olur.

$$\lim_{h\to 0} \vec{E}(P_-) = -\frac{\rho_s \vec{e}_z}{2\varepsilon_0}$$

d) $a \to \infty \to$

$$\lim_{a\to\infty} E(P_+) = \frac{\rho_s \vec{e}_z}{2\varepsilon_0} \lim_{a\to\infty}\left(1-\frac{h}{\sqrt{a^2+h^2}}\right)$$

$$= \frac{\rho_s \vec{e}_z}{2\varepsilon_0}$$

$$\lim_{a\to\infty} \vec{E}(P_-) = -\frac{\rho_s \vec{e}_z}{2\varepsilon_0}$$

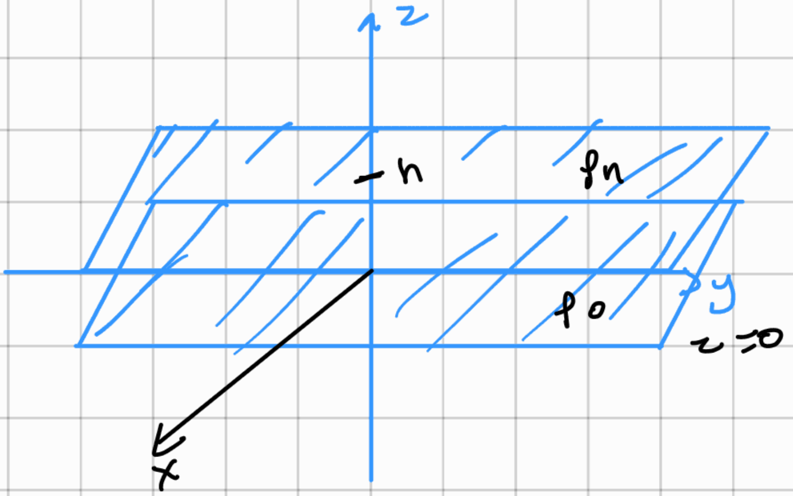
PGS

-5. z = 0 ve z = h düzlemleri üzerinde, sırasıyla, ρ_0 ve ρ_h yoğunlukları ile düzgün yayılı yükler vardır.

a) Pr.-4 ün sonucundan yararlanarak **E** nin açık ifadesini yazınız.

b) $\rho_0 = \rho_h$ halini inceleyiniz.

c) $\rho_0 = -\rho_h$ halini inceleyiniz.



seçtiğimiz noktanın z'si

a) $\vec{E}=?$ $\quad z>h$

$$\vec{E}_I = \vec{E}_0^{+} + \vec{E}_h^{+} = \frac{\rho_0 \vec{e}_z}{2\varepsilon_0} + \frac{\rho_h \vec{e}_z}{2\varepsilon_0}$$

$$= \frac{(\rho_0+\rho_h)\vec{e}_z}{2\varepsilon_0}$$

$$0<z<h \Rightarrow \vec{E}_0^{+} + \vec{E}_h^{-} = \frac{(\rho_0-\rho_h)\vec{e}_z}{2\varepsilon_0}$$

$$z<0 \Rightarrow \vec{E}_0^{-} + \vec{E}_h^{-} = -\frac{(\rho_0+\rho_h)\vec{e}_z}{2\varepsilon_0}$$

b) $\rho_0 = \rho_h$

$$\vec{E}_I = \frac{\cancel{2}\rho_0 \vec{e}_z}{\cancel{2}\varepsilon_0} = \frac{\rho_0 \vec{e}_z}{\varepsilon_0}$$

$$\vec{E}_{II} = 0$$

$$\vec{E}_{III} = \frac{-\rho_0 \vec{e}_z}{\varepsilon_0}$$

c) $\rho_0 = -\rho_h$

$$\vec{E}_I = 0$$

$$\vec{E}_{II} = \frac{\rho_0 \vec{e}_z}{\varepsilon_0}$$

$$\vec{E}_{III} = 0$$